PULLBACK ON PICARD GROUPS IS A BIJECTION ONTO THE SUBGROUP OF FIBRE-TRIVIAL LINE BUNDLES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 21357-21411, immediately following the statement at 21345-21355.

Editorial note. Seven cross-chapter pointers (Algebra, Cohomology of Schemes, Coherent, Cohomology) plus the chapter's own first-order-thickening lemma are recorded in assembly.external_references.

Editorial note. The statement's condition on fibres is used exactly through the vanishing hypothesis; no material is imported from outside the counted proof except as pointers.

Allowed standard prerequisites (editorial).

- ThePicard group of a first order thickening. chapter lemma lemma-picard-group-first-order-thickening, cited in the proof. Provides the infinitesimal step that drives the induction on thickenings.
- Finiteness and descent of finite projective modules. Algebra, lemmas algebra-lemma-finite-projective and algebra-lemma-finite-projective-descends, cited in the proof. Keeps the relevant modules finite projective and lets them be descended along the thickening.
- Cohomology and base change for a proper morphism. Cohomology of Schemes, proposition cohomology-proposition-vanishing-Noetherian and lemma cohomology-lemma-projection-formula, cited in the proof. Vanishing of the first cohomology in the setup and the projection formula for the pushforward of line bundles.
- Formal functions on the stalk and the completion section. Cohomology of Schemes, lemma coherent-lemma-formal-functions-stalk with Algebra, section algebra-section-completion-noetherian, cited in the proof. Controls the completion of the structure sheaf used in the deformation argument.

Source reference. Original reference 00NX: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00O1: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BNH: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02OD: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02O5: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01E8: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02UZ: focused-seeds/geometry/S9-morph-bijection-on-pic/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0C6R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 599–599. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of Noetherian schemes such that $f_*\mathcal{O}_X = \mathcal{O}_Y$, such that the fibres of f have dimension ≤ 1, and such that $H^1(X_y, \mathcal{O}_{X_y}) = 0$ for $y \in Y$. Then $f^* : \mathrm{Pic}(Y) \to \mathrm{Pic}(X)$ is a bijection onto the subgroup of $\mathcal{L} \in \mathrm{Pic}(X)$ with $\mathcal{L}|_{X_y} \cong \mathcal{O}_{X_y}$ for all $y \in Y$.*

Proof. By the projection formula (Cohomology, Lemma 01E8) we see that $f_*f^*\mathcal{N} \cong \mathcal{N}$ for $\mathcal{N} \in \mathrm{Pic}(Y)$. We claim that for $\mathcal{L} \in \mathrm{Pic}(X)$ with $\mathcal{L}|_{X_y} \cong \mathcal{O}_{X_y}$ for all $y \in Y$ we have $\mathcal{N} = f_*\mathcal{L}$ is invertible and $\mathcal{L} \cong f^*\mathcal{N}$. This will finish the proof.

The $\mathcal{O}_Y$-module $\mathcal{N} = f_*\mathcal{L}$ is coherent by Cohomology of Schemes, Proposition 02O5. Thus to see that it is an invertible $\mathcal{O}_Y$-module, it suffices to check on stalks (Algebra, Lemma 00NX). Since the map from a Noetherian local ring to its completion is faithfully flat, it suffices to check the completion $(f_*\mathcal{L})^\wedge_y$ is free (see Algebra, Section 0BNH and Lemma 00O1). For this we will use the theorem of formal functions as formulated in Cohomology of Schemes, Lemma 02OD. Since $f_*\mathcal{O}_X = \mathcal{O}_Y$ and hence $(f_*\mathcal{O}_X)^\wedge_y \cong \mathcal{O}^\wedge_{Y,y}$, it suffices to show that $\mathcal{L}|_{X_n} \cong \mathcal{O}_{X_n}$ for each n (compatibly for varying n. By Lemma 0C6R we have an exact sequence

$$H^1(X_y, \mathfrak{m}_y^n\mathcal{O}_X/\mathfrak{m}_y^{n+1}\mathcal{O}_X) \to \mathrm{Pic}(X_{n+1}) \to \mathrm{Pic}(X_n)$$

with notation as in the theorem on formal functions. Observe that we have a surjection

$$\mathcal{O}_{X_y}^{\oplus r_n} \cong \mathfrak{m}_y^n/\mathfrak{m}_y^{n+1} \otimes_{\kappa(y)} \mathcal{O}_{X_y} \longrightarrow \mathfrak{m}_y^n\mathcal{O}_X/\mathfrak{m}_y^{n+1}\mathcal{O}_X$$

for some integers $r_n \geq 0$. Since $\dim(X_y) \leq 1$ this surjection induces a surjection on first cohomology groups (by the vanishing of cohomology in degrees ≥ 2 coming

from Cohomology, Proposition 02UZ). Hence the H^1 in the sequence is zero and the transition maps $\mathrm{Pic}(X_{n+1}) \to \mathrm{Pic}(X_n)$ are injective as desired.

We still have to show that $f^*\mathcal{N} \cong \mathcal{L}$. This is proved by the same method and we omit the details. □